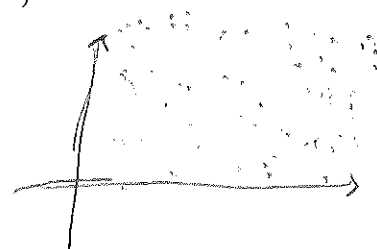
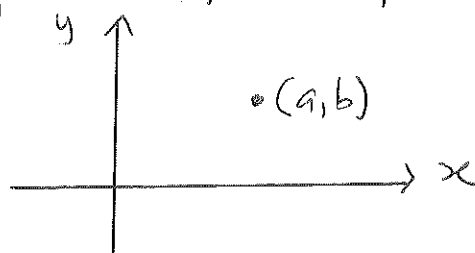


THEME 6.1 : Three-dimensional co-ordinate system

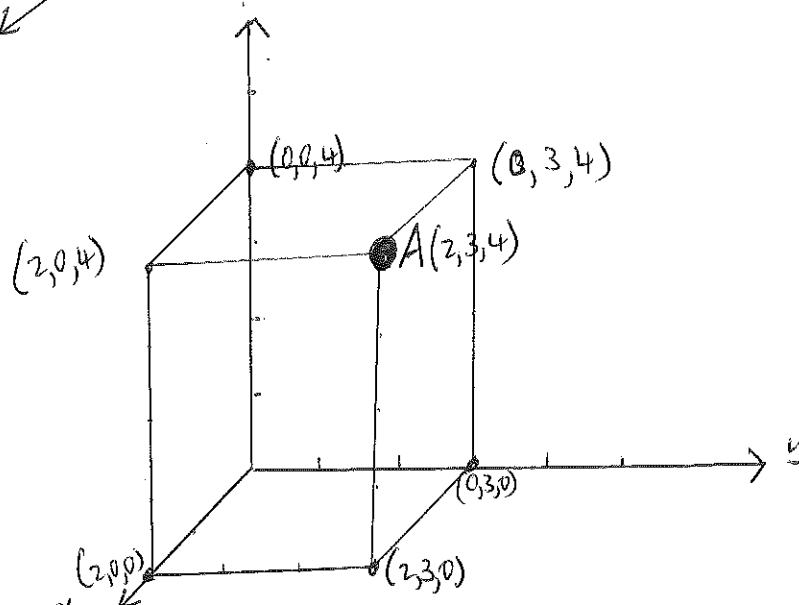
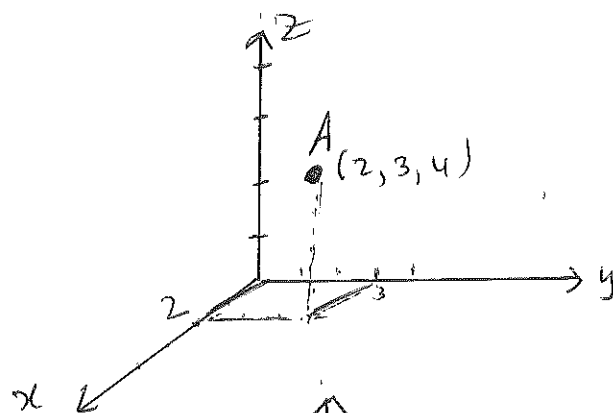
- One dimensional space $\mathbb{R}$ is a line with the points of the form $(a, 0)$

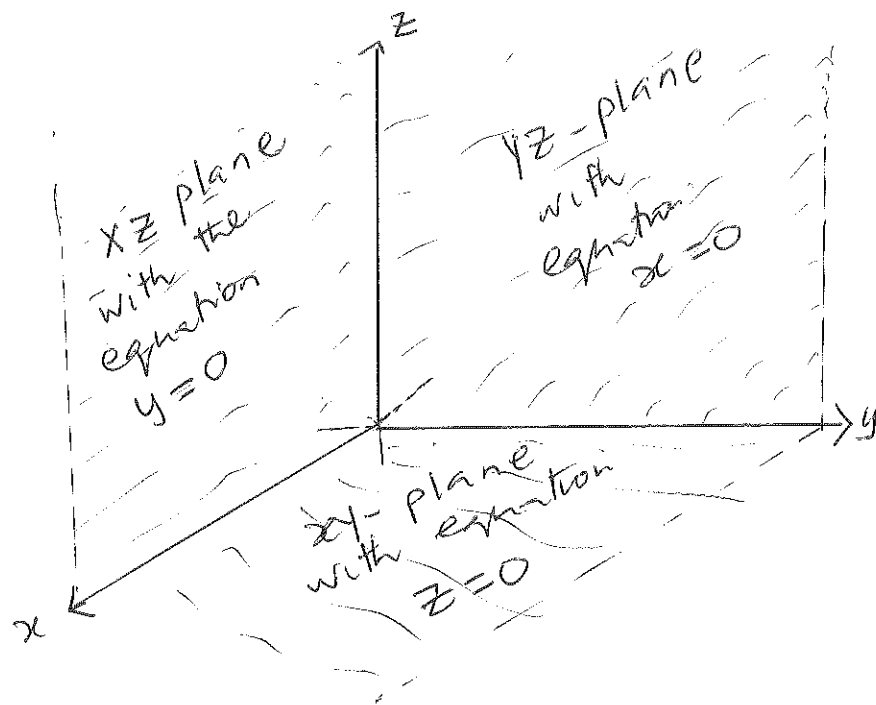


- Two dimensional space $\mathbb{R}^2$ ($\mathbb{R} \times \mathbb{R}$) is a plane with the points of the form (a, b)



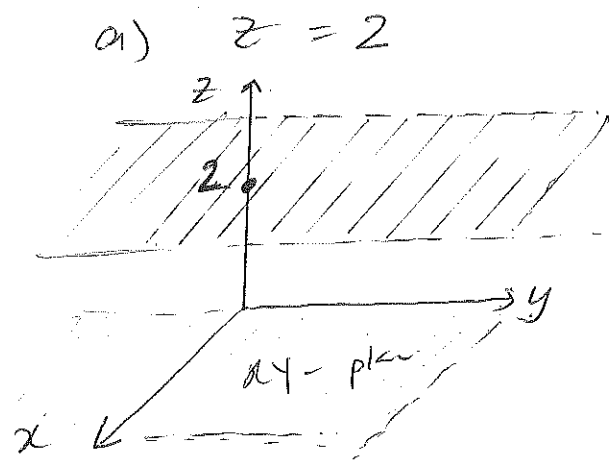
- Three dimensional space $\mathbb{R}^3$ is space with all the points of the form (a, b, c) . Eg. $A(2, 3, 4)$



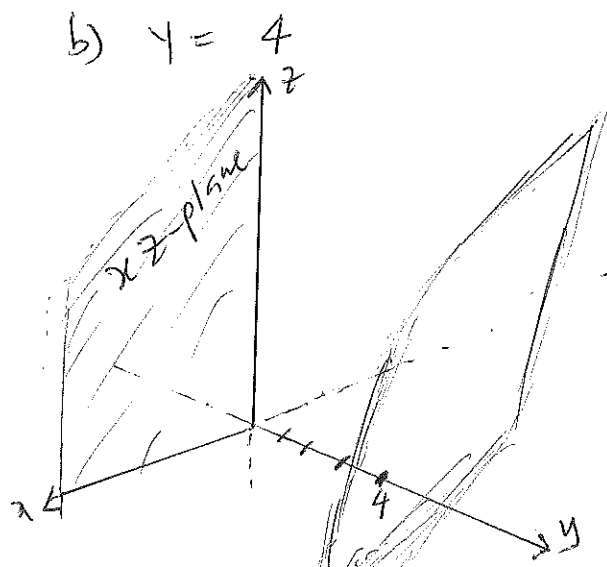


Example: Identify the following surfaces in $\mathbb{R}^3$ represented by

- a) $z=2$
- b) $y=4$
- c) $x=-5$
- d) $x^2+y^2=1$ and $z=3$
- e) $x^2+y^2=1$
- f) $z=y$

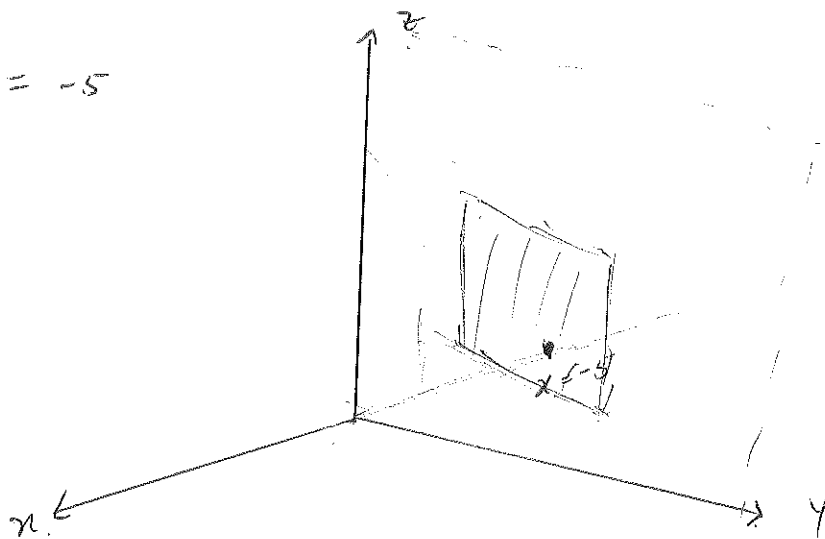


A horizontal plane parallel to the xy -plane and two units above it.



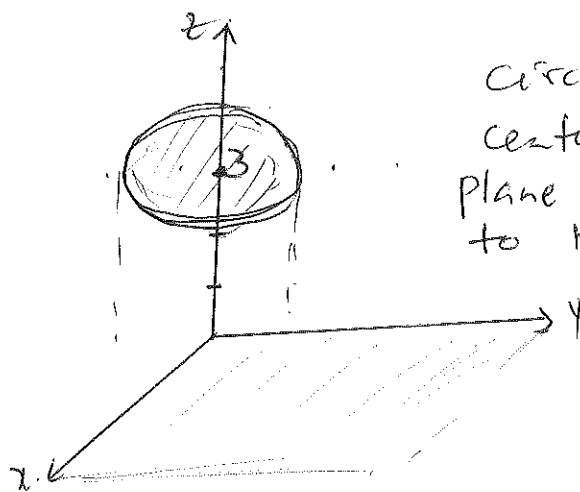
This is a vertical plane to the y -axis but parallel to the xz -plane and 4 units to the right

c) $x = -5$



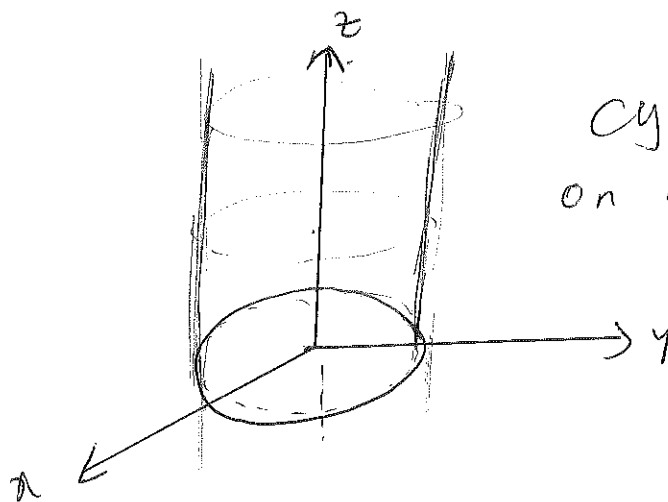
Vertical plane to the x -axis but parallel to the yz -plane and 5 units to the left of it

d) $x^2 + y^2 = 1$, $z = 3$



circle of radius 1
center $(0, 0, 3)$ on the
plane $z = 3$ parallel
to the xy plane.

e) $x^2 + y^2 = 1$



Cylinder
on any $z = k, k \in \mathbb{R}$.

Distance formula in $\mathbb{R}^3$:

distance $|AB|$ between the points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ is

$$|AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

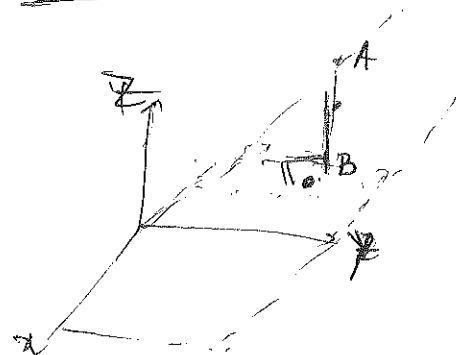
Example: Find the distance between

a) $A(1, 2, -3)$ and $B(-3, 4, 0)$ Solution: $|AB| = \sqrt{29}$

b) $A(-1, 3, 2)$ and the xy -plane.

Solution. Any point on the xy -plane has coordinates $B(-1, 3, 0)$.

The projection of A onto the xy -plane is $B(-1, 3, 0)$



The distance between $A(-1, 2, -3)$ and $B(-1, 3, 0)$ is

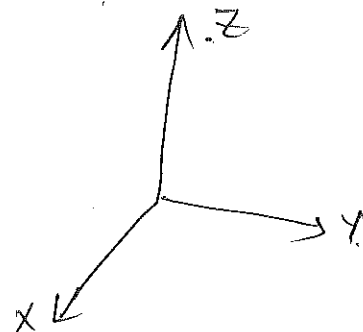
$$|AB| = \sqrt{(-1 - -1)^2 + (3 - 2)^2 + (0 - -3)^2} = \sqrt{(-1 + 1)^2 + (0)^2 + (3)^2} = 3$$

c) Distance between $A(2, -4, 1)$ and the y -axis = ?

Solution. Any point on the y -axis is of the form $(0, y, 0)$. Thus projection of $A(2, -4, 1)$ on $B(0, -4, 0)$ is

of distance

$$|AB| = \sqrt{(2 - 0)^2 + (-4 - -4)^2 + (1 - 0)^2} = \sqrt{5}$$



Example:

Find the centre and radius of the sphere
 $x^2 + y^2 + z^2 + 2x - 6z + 8 = 0$

Solution: Complete the square

$$x^2 + 2x + \left(\frac{2}{2}\right)^2 - \left(\frac{2}{2}\right)^2 + y^2 + z^2 - 6z + \left(\frac{6}{2}\right)^2 - \left(\frac{6}{2}\right)^2 + 8 = 0$$

$$\begin{aligned} \Rightarrow x^2 + 2x + 1 - 1 + y^2 + z^2 - 6z + (-3)^2 - (-3)^2 + 8 &= 0 \\ \Rightarrow x^2 + 2x + 1 + y^2 + z^2 - 6z + (-3)^2 - 9 + 8 &= 0 \\ \Rightarrow x^2 + 2x + 1 - 1 + y^2 + z^2 - 6z + 9 - 9 + 8 &= 0 \\ \Rightarrow x^2 + 2x + 1 + y^2 + z^2 - 6z + 9 &= -8 + 9 + 1 \\ \Rightarrow (x + 1)^2 + (y - 0)^2 + (z - 3)^2 &= 2 = (\sqrt{2})^2 \end{aligned}$$

$\Rightarrow$ centre is $(-1, 0, 3)$ and radius $= \sqrt{2}$

Unit 6.2: Vectors

Definition: A vector is a quantity that has magnitude and direction. Notation: It is often represented by an arrow or a directed line segment.

Example:

Vector from point $A(x_1, y_1, z_1)$ to point $B(x_2, y_2, z_2)$ is written as $\overrightarrow{AB}$ or $\vec{AB}$ and equal to:
 $\overrightarrow{AB} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$

vector $\overline{BA} = \langle x_1 - x_2, y_1 - y_2, z_1 - z_2 \rangle$ (6)

Please note the difference between vectors $\overline{AB}$ and vector $\overline{BA}$

Definition: The vector from the origin $O(0,0)$ to the point $A(x_1, y_1)$ is written as

$$\overrightarrow{OA} = \langle x_1, y_1 \rangle$$

$$\overrightarrow{AO} = -\langle x_1, y_1 \rangle$$

i.e. $\overrightarrow{OA} = \langle x_1, y_1 \rangle = \langle x_1 - 0, y_1 - 0 \rangle$
 $\overrightarrow{AO} = \langle 0 - x_1, 0 - y_1 \rangle$
 $= \langle -x_1, -y_1 \rangle$
 $= -\langle x_1, y_1 \rangle$

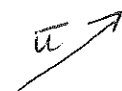
Please note difference in notation between (x_1, y_1) and $\langle x_1, y_1 \rangle$
 point vector

Graphically we can also represent vectors as arrows and also as small letters $\vec{u}, \vec{v}, \vec{w}, \dots$

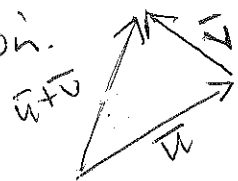


arrow represent the direction of the vector. Length of the arrow represent the magnitude of the vector.

A vector does not have a fixed position



as long as the same magnitude and same direction.



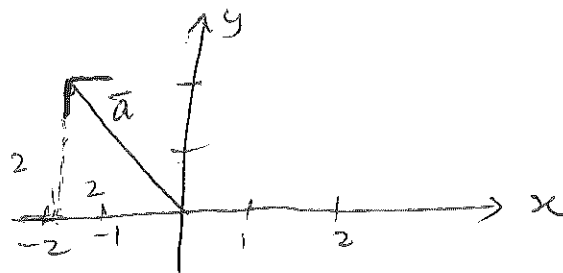
head to tail

Example : sketch vectors :

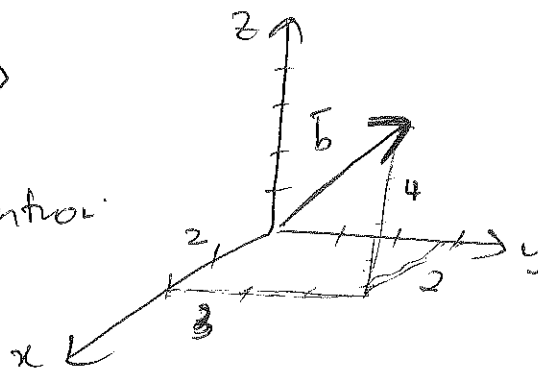
a) $\vec{a} = \langle -2, 2 \rangle$

Solutionlength of the vector
(magnitude) is denoted by

$$|\vec{a}| = \sqrt{(-2)^2 + (2)^2} = \sqrt{4+4} \\ = \sqrt{8}$$



Sketch $\vec{b} = \langle 2, 3, 4 \rangle$

Solutionlength of $\vec{b}$ (ie magnitude) is denoted by

$$|\vec{b}| = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{4+9+16} = \sqrt{29}$$

ie if $\vec{b} = \langle b_1, b_2, b_3 \rangle$, $|\vec{b}| = \sqrt{b_1^2 + b_2^2 + b_3^2}$. If $\vec{a} = \langle a_1, a_2, a_3 \rangle$

$$\vec{b} - \vec{a} = \langle b_1, b_2, b_3 \rangle - \langle a_1, a_2, a_3 \rangle = \langle b_1 - a_1, b_2 - a_2, b_3 - a_3 \rangle$$

(c) Given points $A(3, -2, 0)$, $B(-1, 3, 4)$, $C(2, 0, -1)$

$$\text{Vector } \vec{AB} = \langle -1-3, 3-(-2), 4-0 \rangle \\ = \langle -4, 5, 4 \rangle$$

$$\text{length of } \vec{AB} = |\vec{AB}| = \sqrt{(-4)^2 + 5^2 + 4^2} =$$

$$\text{Vector } \vec{CA} = \langle 3-2, -2-0, 0-(-1) \rangle = \langle 1, -2, -1 \rangle$$

$$\text{vector } \vec{AC} = \langle 2-3, 0+2, -1-0 \rangle = -\langle 1, -2, -1 \rangle$$

Please note the difference between $\vec{AC}$ and $\vec{CA}$

(8)

Properties of vector addition, subtraction and scalar multiplication.

Let $\vec{a}, \vec{b}$ be vectors and c, d be scalars.

1. $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (commutative law for addition)
2. $\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$ (associative law for addition)
3. $\vec{a} + \vec{0} = \vec{a}$ where $\vec{0} = \langle 0, 0, 0, \dots \rangle$ (vector not 0.)
4. $\vec{a} + (-\vec{a}) = \vec{0}$ (as a vector)

5. $c(\vec{a} + \vec{b}) = c\vec{a} + c\vec{b}$
(distributive law for addition)

c is a scalar

6. $(c+d)\vec{a} = c\vec{a} + d\vec{a}$

c and d are scalars.

7. $(cd)\vec{a} = c(d\vec{a})$ (associative law for multiplication)

8. $1\vec{a} = \vec{a}$

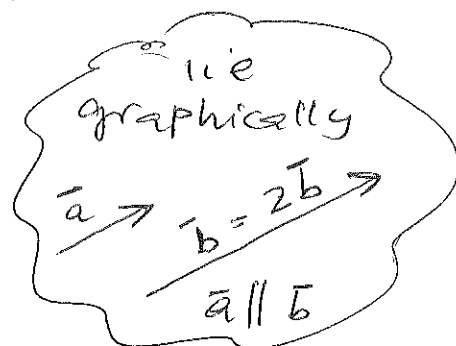
Definition: Vector $\vec{a}$ and $\vec{b}$ are parallel if there exist a $c \in \mathbb{R}$ such that
 $\vec{a} = c\vec{b}$

Example: let the points

$A(0,1)$ $B(1,1)$, $C(0,2)$, $D(3,2)$

be POINTS (not vectors)

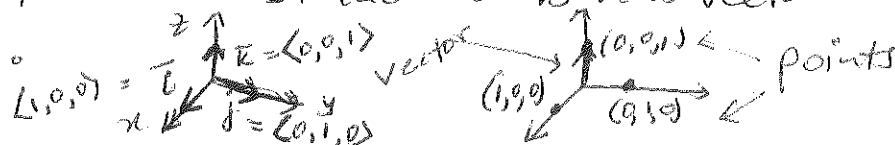
Show that vectors $\vec{AB}$ and $\vec{CD}$ are parallel



Solution: Vector $\vec{AB} = \langle 1-0, 1-1 \rangle = \langle 1, 0 \rangle$
 $\vec{CD} = \langle 3-0, 2-2 \rangle = \langle 3, 0 \rangle$

But $\vec{AB} = \frac{1}{3} \langle 3, 0 \rangle = \frac{1}{3} \vec{CD}$ where
 $c\vec{a} = c \langle a_1, a_2, a_3 \rangle = \langle ca_1, ca_2, ca_3 \rangle$

Vectors $\vec{i}$, $\vec{j}$, $\vec{k}$ are standard basis vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$.



Every vector can be written in terms of these standard basis vectors ($\mathbb{R}^2/\mathbb{R}^3$)

Example: Write vector $\langle -4, -2, 1 \rangle$ in terms of standard basis vectors

Solution: $\langle -4, -2, 1 \rangle = -4\langle 1, 0, 0 \rangle + -2\langle 0, 1, 0 \rangle + 1\langle 0, 0, 1 \rangle$

$$= -4\vec{i} + -2\vec{j} + \vec{k}$$

$$\Rightarrow \langle -4, -2, 1 \rangle = -4\vec{i} - 2\vec{j} + \vec{k}$$

Thus: We can • add and subtract vectors
• scalar multiply vectors

⊗ We have not talked about any "multiplication" nor division of vectors. Stay tuned

⊗ Also any vector $\langle a, b, c \rangle$ can be written as

$$\langle a, b, c \rangle = a\vec{i} + b\vec{j} + c\vec{k}$$

not

$$\langle a, b, c \rangle \neq \langle a\vec{i} + b\vec{j} + c\vec{k} \rangle$$

wrong

⊗ We do not add points

$A(a_1, b_1, c_1)$ $B(a_2, b_2, c_2)$ $C(a_3, b_3, c_3)$

We first form vectors $\vec{AB}$, $\vec{BC}$, $\vec{CB}$, $\vec{BA}$, $\vec{AC}$ etc. Then add vectors

(11)

Please note : Points $A(1, -3, 0)$ $B(-2, 4, 5)$

Vectors $\overrightarrow{AB} = \langle -2-1, 4+3, 5-0 \rangle$
 $= \langle -3, 7, 5 \rangle$

To say $A+B$ is WRONG. It is like adding two points.

We are also going to dot product and cross product VECTORS

NOT POINTS ! ! ! !